



AI APPLICATIONS FOR GLOBAL HEALTH

AI-Assisted Technical Writing: Quick Reference Guide



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1. Prompting for Accuracy

1.1 Prompts That Reduce Hallucinations

Add these instructions to your prompts:

Instruction	Example phrasing
Demand sourcing	Only include information you can attribute to a specific source.
Prohibit invention	Do not make up statistics, citations, or facts. If you don't know, say so.
Request confidence flags	Mark any claim you're uncertain about with [VERIFY].
Limit scope	Base your response only on the documents I've provided.
Break complex tasks	Ask for one section at a time rather than a full report.

Sample system prompt for technical writing:

You are assisting with a technical report for [audience]. Be precise and conservative. Do not invent statistics, citations, or facts. If information is uncertain or outside your knowledge, say "I don't have reliable information on this." Flag claims that require verification with [VERIFY].

1.2 Instructing the Model to Admit Uncertainty

LLMs default to confidence. Override this explicitly:

- "If you're not sure, say so."
- "Distinguish between what you know with high confidence and what you're inferring."
- "Do not guess at numbers or dates."
- "Say 'I don't know' rather than providing uncertain information."

Watch for hedged language that still contains fabrication — phrases like "approximately," "around," or "estimated" may precede invented figures.

1.3 Working with Source Documents

When possible, provide your own sources rather than relying on the model's training data.

How to do it:

- Upload PDFs, Word documents, or data files directly (Claude, ChatGPT, Gemini)

- Paste key excerpts directly into the prompt to ensure the model sees exactly what you need and to focus its attention on the most relevant content
- Provide URLs for tools with web access

Prompt guidance:

- "Use only the attached document(s) to answer."
- "If the answer isn't in the provided materials, say so."
- "Quote directly from the source when possible and note the page or section."

Limitations to know:

- Large documents may be partially processed; ask the model to confirm what it can see
- Scanned PDFs may not be readable; use OCR first if needed
- The model may still draw on background knowledge unless explicitly restricted

2. Getting Current Information

2.1 When and How to Use Web-Enabled Search

LLMs have training data cutoffs, so use web search for any information that may have changed since the last training cutoff date.

Use web search for:

- Current statistics (disease burden, population data, funding figures)
- Recent policy changes or guidelines
- Events from the past 6–12 months
- Verifying whether information has changed since the cutoff

Tools with built-in web search:

- Claude: Enable "Web search" in the interface
- ChatGPT: Enable "Web search" or use the search icon
- Gemini: Web-grounded by default in most modes
- Perplexity: Select "Web" as the source

Prompt tips:

- "Search for the latest WHO guidelines on [topic]."
- "Find current data on [indicator] from [source]."
- "Check whether [fact] is still accurate as of February 2026."
- "Verify this is the most recent version available as of [month/year]."

2.2 Verifying Time-Sensitive Data

Even with web search enabled, verify independently:

- **Check the original source** — AI may summarize incorrectly or pull from outdated pages
- **Note publication dates** — Ensure the data is as current as you need
- **Be specific about dates** — Ask for "2026 data" rather than "recent data"

High-risk time-sensitive content:

- Epidemiological data (incidence, prevalence, mortality)
- Policy and guideline updates
- Funding commitments and disbursements
- Country political situations
- Rapidly changing technological information (AI tools, digital health platforms)

3. Editing vs. Generating

3.1 Using AI to Revise Your Drafts

When you provide your own text for editing, hallucination risk is lower but not eliminated.

Best uses:

- Grammar, spelling, punctuation
- Sentence structure and flow
- Clarity and conciseness
- Tone adjustments for audience

Risks to watch:

- The model may "correct" accurate technical terms to incorrect ones
- Meaning can shift subtly during rewording
- Numbers and names may be inadvertently changed
- Nuance may be lost in simplification

Prompt tips:

- "Edit for clarity only. Do not change technical terms or data."
- "Preserve all numbers and proper nouns exactly as written."
- "Track changes so I can see what you modified."

3.2 Using AI to Generate New Text

Generating text from scratch carries higher risk. The model draws on training data, which may be outdated, incorrect, or fabricated.

Higher-risk tasks:

- Writing background or literature review sections
- Summarizing evidence you haven't provided
- Describing methods or results you haven't specified
- Making claims about organizations, policies, or data

Risk mitigation:

- Provide detailed outlines, bullet points, or source documents
- Generate in small sections rather than full documents
- Treat all generated content as a first draft requiring verification
- Verify factual claims and citations

3.3 Executive Summaries and Abstracts

These high-visibility sections are often generated or condensed by AI. Errors here shape how readers understand the entire document.

Common problems:

- Overgeneralization that misrepresents findings
- Key caveats or limitations dropped
- Numbers rounded or changed
- Conclusions stronger than the evidence supports

Verification checklist:

- Every claim in the summary is supported by the main document
- Numbers match exactly
- Caveats and limitations are preserved
- No new information has been introduced

4. Verifying High-Risk Content

4.1 Data and Statistics

LLMs frequently fabricate or misremember numbers. Treat every statistic as unverified until confirmed.

Always verify:

- Prevalence, incidence, and mortality rates
- Population figures
- Program coverage and reach
- Costs, budgets, and funding amounts
- Percentages and ratios
- Dates and timeframes

How to verify:

- Check the original source
- Confirm the year and geographic scope of the data
- Match the exact indicator definition (e.g., "under-5 mortality" vs. "infant mortality")

Red flags:

- Round numbers (e.g., "approximately 1 million") — may be invented
- Precise figures without a source — likely fabricated
- Numbers that seem plausible but can't be traced

4.2 Technical Terminology

Global health has precise terminology. LLMs may use outdated terms, incorrect definitions, or non-standard language.

Check for:

- WHO and agency-specific definitions
- ICD classification codes
- Standardized intervention names
- Correct acronym expansions
- Current vs. deprecated terminology (e.g., "LMIC" vs. older classifications)

Verify against:

- WHO glossaries and fact sheets

- Disease-specific program guidelines
- Agency style guides (UNAIDS, UNICEF, etc.)

4.3 Tables and Structured Data

LLMs make frequent errors in tables: misaligned rows, incorrect totals, duplicated or missing cells, fabricated data.

Verification checklist:

- Row and column totals add up correctly
- Data matches the cited source exactly
- No duplicated or missing entries
- Headers accurately describe the content
- Units are consistent throughout

4.4 Quotations and Paraphrasing

LLMs may fabricate quotes or subtly distort meaning when paraphrasing.

For direct quotes:

- Verify the exact wording against the original source
- Confirm the speaker/author attribution
- Check context — quotes may be accurate but misleading out of context

For paraphrasing:

- Compare the AI's paraphrase to the original passage
- Check that the meaning is preserved
- Watch for oversimplification that changes the message

Prompt tip: "Quote directly from the source and provide the page number" — then verify.

4.5 Country and Context Sensitivity

LLMs can get political and cultural details wrong in ways that damage credibility or cause offense.

Verify:

- Country and territory names (use UN or publisher conventions)
- Disputed borders and regions — check your publisher's or funder's style
- Current political leadership and government names
- Culturally appropriate framing and language
- Sensitive historical events

Resources:

- UN terminology database (UNTERM)
- Publisher or funder style guides
- Country-specific UN or government sources

5. Document Consistency

5.1 Terminology and Acronyms

LLMs may use terms inconsistently, especially across long documents or multiple prompts.

Common problems:

- Switching between synonyms (e.g., "health workers" vs. "healthcare workers")
- Inconsistent acronym expansion (defining the same acronym multiple times, or not at all)
- Mixing British and American spelling
- Using different names for the same organization or program

Prevention:

- Provide a terminology list in your prompt: "Use these terms consistently: [list]"
- Specify spelling convention: "Use British spelling throughout"
- Define acronyms once: "Expand acronyms on first use only"

Checking:

- Search the document for key terms and verify consistency
- List all acronyms and confirm each is defined once
- Use find/replace to standardize variations

5.2 Contradictions Across Sections

LLMs can contradict themselves, especially when generating sections separately or in long conversations.

Common contradictions:

- Different numbers for the same indicator
- Conflicting descriptions of methods or timelines
- Inconsistent framing of findings or recommendations
- Contradictory statements about scope or limitations

Prevention:

- Generate documents in sequence, providing prior sections as context
- Summarize key facts and decisions at the start of each prompt
- Ask the model to review for internal consistency: "Check this document for any contradictions"

Checking:

- Read the full document through, noting key claims
- Cross-check numbers and dates across sections
- Compare introduction/abstract with conclusions

5.3 Checking Long Documents

LLM context windows have limits. For long documents, the model may lose track of earlier content.

Strategies during writing:

- **Work in one conversation** — Keep all your work in a single thread so the model retains earlier context.
- **Use Projects or custom instructions** — Claude Projects, ChatGPT custom GPTs, or Gemini Gems let you store key reference information that persists across conversations.
- **Keep a "key facts" list** — Maintain a short list of critical numbers, terms, and decisions. Paste it at the start of new prompts.

After completing a draft:

- **Final consistency review** — Upload the complete draft and ask: "Review this document for internal contradictions and inconsistent terminology."

6. Citations & References

LLMs routinely fabricate citations — inventing authors, titles, journals, DOIs, and URLs. This section provides a workflow for gathering evidence, drafting with citations, and verifying accuracy.

6.1 Research and Evidence Gathering

Academic Literature (Elicit)

- **Define your research questions.** Break your document into sections, each with a core question.
- **Search Elicit for each question.** Review the papers and summaries it returns.
- **Build a table of evidence.** For each useful finding, record the claim, source details, exact quote, and link.
- **Download source PDFs** for citations you plan to use.
- **Download your evidence table** from Elicit (as CSV or copy/paste).

Reports and Other Non-Journal Sources

- **Search for reports, guidelines, and other documents** using Google, an LLM with web search, or other tools. Before adding any source to your evidence, verify the document exists, the link works, and it's the current version.
- **Download and organize the PDFs** using a consistent naming format.
- **Extract key claims using an LLM.** Upload each PDF and ask:
“Extract the key findings, statistics, and recommendations from this document. For each, provide the exact quote and page number.”
- **Add to your evidence table:** claim, source details, exact quote, page number, and URL.

6.2 Drafting

- **Combine your evidence** from academic and non-journal sources into one document, organized by section or theme.
- **Give the LLM your evidence and instructions:**
“Write [document type] on [topic]. Use the following evidence. Cite each source where indicated. Do not add claims or citations beyond what I’ve provided.”
- **Draft section by section** for longer documents.
- **Review for flow and completeness.** Focus on argument and structure. Citation verification comes next.

6.3 Citation Verification

- **List every factual claim and its citation.** Ask the LLM to extract these as a numbered list.

- **Sort claims by risk:**
 - **High risk:** Statistics, cause-and-effect claims, policy recommendations, central arguments
 - **Medium risk:** General findings with specific sources
 - **Low risk:** Well-established facts, claims with multiple sources
- **Verify high-risk claims manually.** Open the source and read the relevant section yourself.
- **Verify medium-risk claims using an LLM.** Upload the source PDF and ask:
“I want to cite this document to support the following claim: [claim]. Does the source support this? Quote the specific passage.”
- **Spot-check low-risk claims** if time allows.

6.4 Final Review

- **Check consistency.** Make sure any revised claims still fit the surrounding text.
- **Check your reference list.** Every in-text citation should appear in references and vice versa.
- **Check formatting** against the required style guide.

Never submit a document with unverified AI-generated citations.

7. Formatting & Style Compliance

7.1 Working with Style Guides

LLMs don't reliably follow specific style guides unless explicitly instructed.

Common style requirements:

- UN style (UN Editorial Manual)
- Academic journal styles (Lancet, BMJ, JAMA, etc.)
- Donor templates (Global Fund, USAID, Gates Foundation)
- Organizational house styles

How to use:

- Upload the style guide or template if available
- Specify key requirements in your prompt: "Follow UN Editorial Manual style. Use British spelling. Write dates as 1 January 2026."
- Provide examples of correctly formatted content
- Ask the model to format a sample section first, review it, then proceed

7.2 Common Formatting Errors

Element	Common LLM errors
Dates	Inconsistent formats (1/1/26 vs. 1 January 2026 vs. January 1, 2026)
Numbers	Inconsistent use of commas, decimals, and thousands separators
Percentages	Mixing "percent" and "%"; inconsistent decimal places
Citations	Wrong format for the required style
Headings	Inconsistent capitalization (title case vs. sentence case)
Lists	Inconsistent punctuation and capitalization
Spelling	Mixing British and American conventions
Units	Inconsistent formatting (kg vs. kilograms vs. KG)

Checking:

- Review a sample of each element type before finalizing
- Use find/replace to check for common inconsistencies
- Run a final manual pass focused only on formatting

8. Quality Assurance Checklist

8.1 Fact-Checking Protocol

Before finalizing any AI-assisted document, verify:

Statistics and data:

- ☐ Every number traced to an original source
- ☐ Year and geographic scope confirmed
- ☐ Indicator definitions match exactly
- ☐ Calculations checked independently

Citations and references:

- ☐ Every citation verified to exist
- ☐ Author, title, year, and publication details confirmed
- ☐ DOIs/URLs resolve correctly
- ☐ Source content supports the claim being made

Technical content:

- ☐ Terminology matches current standards
- ☐ Organization and program names are correct
- ☐ Country and context details verified
- ☐ Acronyms defined correctly and consistently

Quotations:

- ☐ Exact wording verified against original
- ☐ Attribution confirmed
- ☐ Context preserved

8.2 Pre-Submission Verification Steps

Document review:

- ☐ Read the full document for contradictions
- ☐ Cross-check numbers across sections (abstract, body, tables)
- ☐ Verify all claims in executive summary match the main text
- ☐ Confirm terminology and acronyms are consistent throughout

Formatting check:

- ☐ Style guide requirements met
- ☐ Dates, numbers, and units formatted consistently
- ☐ Spelling convention consistent throughout
- ☐ Headings and lists follow required style





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